

## ① Slip layer

↳ below yield stress, the bulk material moves as a rigid body  
slip thickness

① Plot  $\rightarrow$  (shear stress - residual stress) vs strain rate

② Slip thickness = 
$$\frac{\text{viscosity of silicon oil} \times \text{gap b/w plates}}{\text{slope from ①}}$$

## ② Shear Thickening

↳ viscosity increases with increase in shear rate

shear thickening  $\rightarrow$  more pronounced at higher concentrations  
 $\downarrow$   
more viscosity increase with shear rate

## ③ Static & Dynamic Yield Stress Analysis

yield stress  $\rightarrow$  min stress for a material to undergo a transition from a solid like behaviour to more fluid like (exhibit deformation or flow)

no yield stress for  $\rightarrow$  Newtonian fluids

$\sigma = k(\dot{\gamma}) \rightarrow$  Newtonian fluid

$\sigma_y \rightarrow$  residual stress

$\sigma = \sigma_y + k(\dot{\gamma}) \rightarrow$  Bingham fluid

$\sigma = \sigma_y + k(\dot{\gamma})^n \rightarrow$  Herschel-Bulkley fluid

$\sigma = k(\dot{\gamma})^n \rightarrow$  Power law  $\rightarrow$  used to describe non-Newtonian behaviour

Dynamic yield stress  $\rightarrow$  we go from high shear rate to low shear rate  
 static " " " " " low " " to high " "

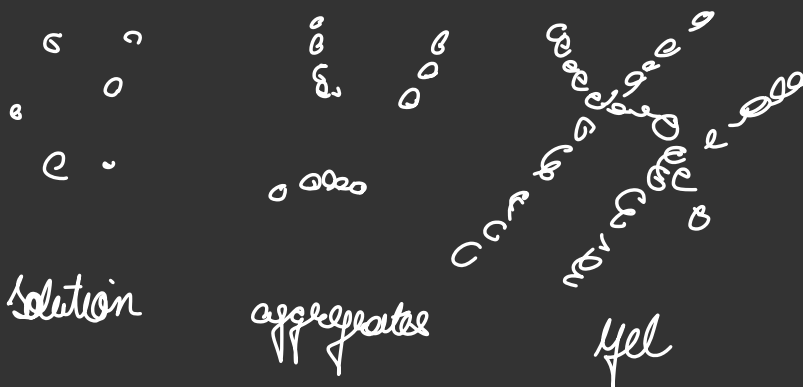
$E \uparrow$  Dynamic yield stress  $\uparrow$  and static yield stress  $\uparrow$

For same electric field  $\rightarrow$  Dynamic Yield stress  $>$  static yield stress

$E \uparrow$  Viscosity  $E \uparrow$

Increase in conc. leads to higher yield stress (beg of more solid like behaviour)

#### ④ Rheology of milk products



$\rightarrow$  semi solid materials with a continuous solid like structure  
 so exhibit viscoelastic behaviour  $\leftarrow$  & significant liquid like component

Curd  $\rightarrow$  liquid (whey) trapped in protein network

Butter  $\rightarrow$  multi phase system (presence of solid & liquid phases within its structure)  
 $\rightarrow$  also contains amorphous phase

### ③ physical ageing of carboxyl

$G_1'$  → shows weak change as a fun. of time

creep test → shows a steady change in behaviour  
(more for 3 mins)

### ④ Raw data for oscillatory rheology

$$G_1' = \frac{\sigma_0 \cos \phi}{\gamma_0}$$

$$G_1'' = \frac{\sigma_0 \sin \phi}{\gamma_0}$$

### ⑦ gel point rheology → protein undergoes a phase change from sol to gel

gel → 3D network structure of polymers dispersed in liquid phase

↳ high viscosity, 3D network structure

gelation → process in which macromolecular polymer chains are connected together by chemical / physical bonds

gel point → storage modulus of fluid exceeds loss modulus

↓

system loses its fluidity

determined by observing a sudden rise in viscosity & transition to a more solid like state

at gel point → loss factor becomes independent of frequency

↓

so all  $G_1''$  of diff.  $f$  pass through same pt on the  $G_1''$  vs time graph

## ⑧ Effect of Concentration on Rheology

Einstein Equation  $\rightarrow$  dilute, Newtonian fluid.

↓  
doesn't function  
at high conc.

↓  
particle -  
particle interactions  
are negligible

$$\eta = \eta_0 (1 + 2.5\phi)$$

$\hookrightarrow$  initial bulk viscosity  $\quad \hookrightarrow$  volume fraction of dispersed component

Kreger Dougherty Eqn  $\rightarrow$  Non-Newtonian fluids, particles interactions present

$\hookrightarrow$  not accurate for dilute solns

$$\eta = \eta_0 \left( 1 - \frac{\phi}{\phi_m} \right)^{-2.5\phi_m}$$

$\hookrightarrow \phi_m \rightarrow$  "close packing" fraction at which viscosity becomes infinite

## ⑨ p-H induced sol-gel transition

$\hookrightarrow$  pH can affect charge & solubility of particles in a sol leading to aggregation & gelation.

↓  
Process involves ionizing specific groups within polymer chains

## ⑩ Analysis of aging of foam

Foam  $\rightarrow$  pockets of gas trapped in liquid / solid medium

Physical aging  $\rightarrow$  gradual structure rearrangement of polymer chains even below  $T_g$  leading to observable increase in material stiffness & decrease in its ability to flow.

Aging time  $\uparrow$  Relaxation Time  $\uparrow$   
.. ..  $\uparrow$  storage modulus  $\uparrow$

## ⑪ DLS

$$g_2(q, \tau) = \frac{\langle I(q, t), I(q, t + \tau) \rangle}{\langle I(q, t)^2 \rangle}$$

$$g_2(q, \tau) = 1 + B \underbrace{|g_1(q, \tau)|^2}_{\text{electric field correlation function}}$$

$$g_1(q, \tau) = \frac{\langle E(q, \tau) E(q, t + \tau) \rangle}{\langle |E(q, \tau)|^2 \rangle}$$

For a mono-disperse  $g_1(q, \tau) = \exp(-\Gamma \tau)$

$$\Gamma = q^2 D$$

$$D = \frac{kT}{6\pi\eta R}$$

$$q = \frac{4n\pi}{\lambda} \sin\left(\frac{\theta}{2}\right) \quad n \rightarrow \text{refractive index}$$

$$g_2(\tau) = B[1 + B|g_1(\tau)|^2]$$

For monodisperse

$$g_1(\tau) = \exp(-\Gamma\tau)$$

$$g_2(\tau) = B[1 + B\exp(-2\Gamma\tau)]$$

$$\Gamma = Dq^2$$

$$\ln \frac{g_2(\tau) - B}{B} = \ln B - 2\Gamma\tau$$

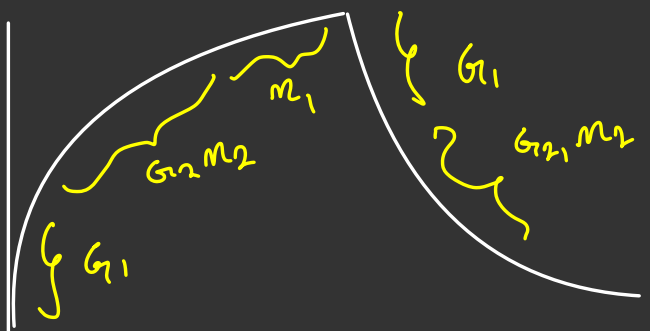
$$\ln \frac{g_2(\tau) - B}{B} = \ln B - 2\underset{\substack{\downarrow \\ \text{avg decay}}}{\Gamma}\tau + \frac{k_2^2}{2}\tau^2 - \frac{k_3^3}{3}\tau^3 + \frac{k_4^4}{12}\tau^4$$

$$g_1(\tau) = \sum_i' g_{1i}(\tau) e^{-\Gamma_i \tau}$$

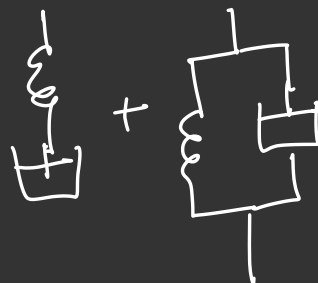
intensity weighted coefficient

## (12) ER fluids

1. Creep recovery at diff. electric fields
2. stress relaxation exp. " "



Maxwell + Kelvin-Voigt



$E \uparrow$ , solid-like behavior  $\uparrow$

Relaxation time scale  $\uparrow$

more time to relax

